

An Extremal Eigenvalue Problem*

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1. Introduction

We consider the problem

$$(1.1) \quad y'' + \lambda \phi(x)y = 0, \quad y = 0 \quad \text{at} \quad x = \pm \frac{1}{2},$$

where $\phi(x)$ is piecewise continuous with a bounded number of jumps, and satisfying the inequalities

$$(1.2) \quad 0 < \alpha \leq \phi(x) \leq 1, \quad -\frac{1}{2} \leq x \leq \frac{1}{2}.$$

Among all such functions $\phi(x)$ we determine which ones will provide extreme values (max and min) for the ratio λ_2/λ_1 , where λ_1 and λ_2 are the first and second eigenvalues, respectively, of (1.1). We also consider extremizing λ_j/λ_k under the same conditions.

Keller [1] investigated the problem of minimizing λ_2/λ_1 , over the same class of functions. Using the conjecture that the minimizing function was symmetric about $x=0$ and had two jumps, he determined the minimizing function and the minimum ratio. In this paper, we prove that the extremizing function in both cases is symmetric about $x=0$ and has two jumps. The maximizing function, and the maximum ratio, are also determined.

We note that treating the problem on the interval $-l < x < l$, and requiring the function $\phi(x)$ to satisfy $0 < a < \phi(x) \leq A$ does not alter the results.

2. Preliminaries

Let P_α denote the class of piecewise continuous functions, with a bounded number of jumps, which satisfy the inequalities (1.2). The eigenvalues λ_1 and λ_2 are continuous functions of ϕ , and λ_1 is positive and bounded away from zero. Hence, λ_2/λ_1 is a continuous function of ϕ for ϕ in P_α with $\alpha > 0$. If we restrict P_α to those functions which have at most two jumps, the resulting class is compact, and λ_2/λ_1 takes on both a maximum and a minimum value. We shall prove that the extremizing function over all P_α has at most two

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jumps, so that λ_2/λ_1 takes on a maximum, and a minimum, value within P_α . These extreme values are denoted by

$$(2.1) \quad \mu(\alpha) = \min_{\phi \in P_\alpha} \frac{\lambda_2(\phi)}{\lambda_1(\phi)},$$

$$(2.2) \quad \nu(\alpha) = \max_{\phi \in P_\alpha} \frac{\lambda_2(\phi)}{\lambda_1(\phi)}.$$

If $\alpha_1 \leq \alpha_2$, we easily see that

$$(2.3) \quad \mu(\alpha_1) \leq \mu(\alpha_2),$$

$$(2.4) \quad \nu(\alpha_1) \geq \nu(\alpha_2).$$

It also follows easily that

$$(2.5) \quad 1 < \mu(\alpha) \leq \mu(1) = 4,$$

$$(2.6) \quad 4 = \nu(1) \leq \nu(\alpha) < \infty, \quad 0 < \alpha \leq 1.$$

The above also applies to $\mu_{j,k}(\alpha) = \min \lambda_j/\lambda_k$ and $\nu_{j,k}(\alpha) = \max \lambda_j/\lambda_k$, except that $\mu_{j,k}(1) = \nu_{j,k}(1) = j^2/k^2$.

3. Properties of the Extremizing Function

We present now the main results concerning the structure of the extremizing function. The following theorem is proved in [1]:

THEOREM 0. *Any extremizing function is piecewise constant, taking on the values α and 1.*

The proofs of the remaining theorems are given in Section 5.

THEOREM 1. *Any extremizing function of $\phi_0(x)$ has either no jumps or precisely two jumps.*

THEOREM 2. *Any extremizing function $\phi_0(x)$ is symmetric about $x = 0$.*

In the case of maximization, ϕ_0 is determined by (4.1)–(4.5). Since we do not know that these equations provide a unique pair of jump points, we use the phrase “any extremizing function” in Theorems 1 and 2. The theorems

show that the extremizing function for λ_2/λ_1 is either a constant or has two jumps and is symmetric about $x=0$. If ϕ_0 is a constant, $\lambda_2/\lambda_1=4$. Since we know that $\mu(\alpha)\leq 4\leq \nu(\alpha)$ for $0<\alpha\leq 1$, we can assume that the extremizing function is not a constant, and thus it has two symmetric jumps.

In order to state our final theorem, we need a definition. Let $\phi_0(x)$ extremize λ_2/λ_1 , and define

$$(3.1) \quad \phi_n(x) = \phi_0(n(x + \tfrac{1}{2}) - \tfrac{1}{2}), \quad -\tfrac{1}{2} \leq x \leq -\tfrac{1}{2} + 1/n, \quad n = 2, 3, \dots$$

Extend $\phi_n(x)$ periodically, with period $1/n$, to $-\tfrac{1}{2} \leq x \leq \tfrac{1}{2}$. With this definition, $\phi_n(x)$ belongs to P_α whenever ϕ_0 does.

THEOREM 3. *If $\phi_0(x)$ extremizes λ_2/λ_1 , then $\phi_n(x)$ extremizes λ_{2n}/λ_n in the same sense.*

4. Determination of ϕ_0 , λ_1 and λ_2 for a Maximum

We now solve the problem of maximizing the ratio λ_2/λ_1 . Using arguments similar to Keller's, the following equations can be derived for determining x_0 , the positive jump point of ϕ_0 , λ_1 , and λ_2 :

$$(4.1) \quad \sin(\alpha\lambda_1)^{1/2}(\tfrac{1}{2} - x_0) \sin \lambda_1^{1/2} x_0 = \alpha^{1/2} \cos \lambda_1^{1/2} x_0 \cos(\alpha\lambda_1)^{1/2}(\tfrac{1}{2} - x_0),$$

$$(4.2) \quad \sin(\alpha\lambda_2)^{1/2}(\tfrac{1}{2} - x_0) \cos \lambda_2^{1/2} x_0 = -\alpha^{1/2} \sin \lambda_2^{1/2} x_0 \cos(\alpha\lambda_2)^{1/2}(\tfrac{1}{2} - x_0),$$

$$(4.3) \quad N_1 \sin(\alpha\lambda_1)^{1/2}(\tfrac{1}{2} - x_0) \cos \lambda_1^{1/2} x_0 = N_2 \sin(\alpha\lambda_2)^{1/2}(\tfrac{1}{2} - x_0) \sin \lambda_2^{1/2} x_0,$$

where the normalization constants N_1 and N_2 are given by

$$(4.4) \quad N_1^{-2} = \sin^2(\alpha\lambda_1)^{1/2}(\tfrac{1}{2} - x_0) \left[x_0 + \frac{\sin 2\lambda_1^{1/2} x_0}{2\lambda_1^{1/2}} \right] \\ + \alpha \cos^2 \lambda_1^{1/2} x_0 \left[\tfrac{1}{2} - x_0 - \frac{\sin 2(\alpha\lambda_1)^{1/2}(\tfrac{1}{2} - x_0)}{2(\alpha\lambda_1)^{1/2}} \right],$$

$$(4.5) \quad N_2^{-2} = \sin^2(\alpha\lambda_2)^{1/2}(\tfrac{1}{2} - x_0) \left[x_0 - \frac{\sin 2\lambda_2^{1/2} x_0}{2\lambda_2^{1/2}} \right] \\ + \alpha \sin^2 \lambda_2^{1/2} x_0 \left[\tfrac{1}{2} - x_0 - \frac{\sin 2(\alpha\lambda_2)^{1/2}(\tfrac{1}{2} - x_0)}{2(\alpha\lambda_2)^{1/2}} \right].$$

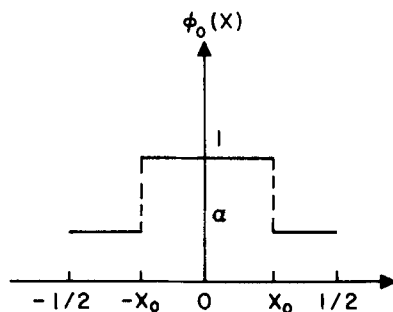


Figure 1. The function $\phi_0(x)$ which maximizes λ_2/λ_1 in the class P_α of piecewise-continuous functions $\phi(x)$ with $\alpha \leq \phi(x) \leq 1$.

We solved the system (4.1)–(4.5) numerically for values of α with $10^{-4} \leq \alpha \leq 1$. For small values of α , we determined the following asymptotic expansions for x_0 , λ_1 , λ_2 and λ_2/λ_1 :

$$(4.6) \quad x_0 \sim \frac{1}{4}\alpha^{1/2} + O(\alpha),$$

$$(4.7) \quad \lambda_1 \sim \frac{8}{\alpha^{1/2}} + O(1),$$

$$(4.8) \quad \lambda_2 \sim \frac{4\pi^2}{\alpha} - \frac{8\sqrt{2}\pi}{\alpha^{3/4}} + O(\alpha^{-1/2}),$$

$$(4.9) \quad \max \frac{\lambda_2}{\lambda_1} = \nu(\alpha) \sim \frac{\pi^2}{2\alpha^{1/2}} + O(\alpha^{-1/4}).$$

The graph of the maximizing functions is given in Figure 1. Figures 2 and 3 give graphs of x_0 and $\nu(\alpha)$ as functions of α .

5. Proofs

THEOREM 1. *Any extremizing function has either no jumps or two jumps.*

Proof: We first note that the ratio λ_2/λ_1 is the same for the problems

$$(5.1) \quad \frac{d^2 y}{dx^2} + \lambda \phi(x)y = 0, \quad y(x_0) = y(x_1) = 0,$$

$$(5.2) \quad \frac{d^2 y}{dx^2} + \lambda \phi(L(x))y = 0, \\ y(L^{-1}(x_0)) = y(L^{-1}(x_1)) = 0,$$

where $L(x) \equiv cx + d$, with $c \neq 0$, d , x_0 and x_1 arbitrary real numbers. Using this observation, we now prove that for each fixed α , $0 < \alpha \leq 1$, the identities $\nu_{2n,n}(\alpha) = \nu(\alpha)$ and $\mu_{2n,n}(\alpha) = \mu(\alpha)$ hold for $n = 1, 2, \dots$.

LEMMA 1. $\mu_{2n,n}(\alpha) \leq \mu(\alpha)$ and $\nu_{2n,n}(\alpha) \geq \nu(\alpha)$ for $0 < \alpha \leq 1$ and $n = 1, 2, \dots$.

Proof: Let $\phi_0(x)$ extremalize λ_2/λ_1 and let $\phi_n(x)$ be defined as in (3.1). Let $\tilde{y}_n(x)$ and $\tilde{y}_{2n}(x)$ denote the n -th and $2n$ -th eigenfunctions of (1.1) with

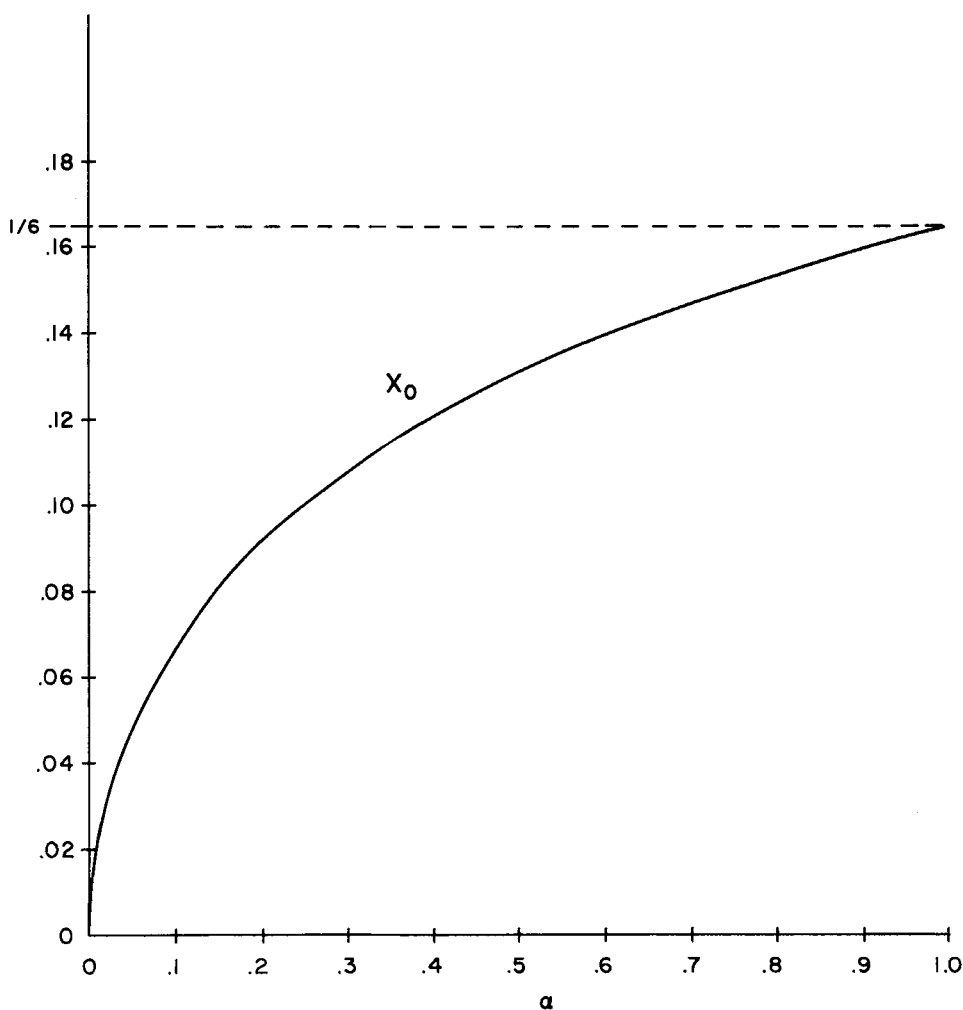


Figure 2. The point of discontinuity x_0 of the maximizing function $\phi_0(x)$, as a function of α .

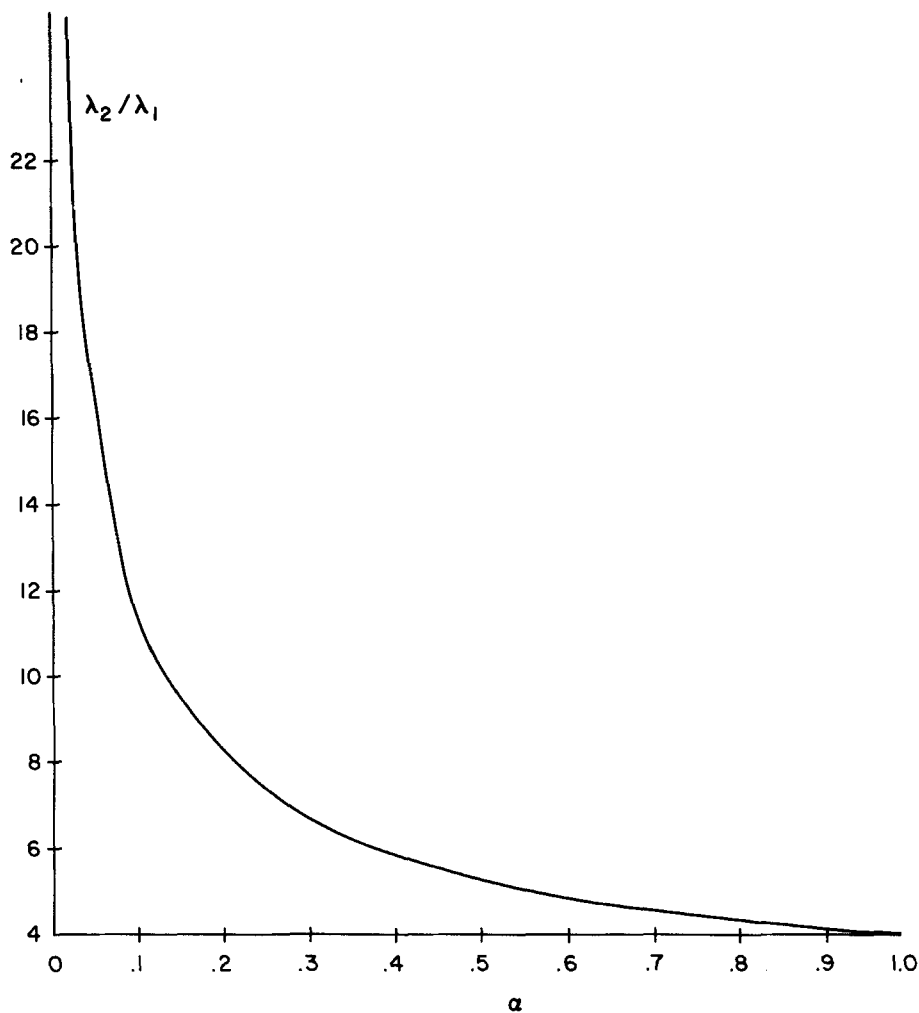


Figure 3. The maximum value $\nu(\alpha)$ of the ratio λ_2/λ_1 as a function of α .

$\phi(x)$ taken as $\phi_n(x)$, with corresponding eigenvalues $\tilde{\lambda}_n$ and $\tilde{\lambda}_{2n}$. It follows that

$$\tilde{y}_n(x) = M_1(-s)^j y_1 \left(n \left(x + \frac{1}{2} - \frac{j}{n} \right) - \frac{1}{2} \right), \quad (5.3)$$

$$\tilde{y}_{2n}(x) = M_2 t^j y_2 \left(n \left(x + \frac{1}{2} - \frac{j}{n} \right) - \frac{1}{2} \right), \quad (5.4)$$

$$-\frac{1}{2} + \frac{j}{n} \leq x \leq -\frac{1}{2} + \frac{j+1}{n}, \quad j = 0, 1, \dots, (n-1),$$

where $s = |y_1'(\frac{1}{2})/y_1'(-\frac{1}{2})|$ and $t = |y_2'(\frac{1}{2})/y_2'(-\frac{1}{2})|$. We call s and t the transmission coefficients of y_1 and y_2 , respectively. M_1 and M_2 are normalization constants.

Let $L(x) = n(x + \frac{1}{2}) - \frac{1}{2}$, $x_0 = -\frac{1}{2}$ and $x_1 = \frac{1}{2}$. We have, by this definition, $L^{-1}(x_0) = -\frac{1}{2}$ and $L^{-1}(x_1) = -\frac{1}{2} + 1/n$. By the remark above, the ratio λ_2/λ_1 is the same for the following problems:

$$(5.5) \quad y'' + \lambda \phi_0(x)y = 0, \quad y(-\frac{1}{2}) = y(\frac{1}{2}) = 0,$$

$$(5.6) \quad y'' + \lambda \phi_0(L(x))y = 0, \quad y(-\frac{1}{2}) = y(-\frac{1}{2} + 1/n) = 0.$$

We observe that problem (5.5) is problem (1.1) with $\phi = \phi_0$, and that $\phi_0(L(x)) = \phi_n(x)$ for $-\frac{1}{2} \leq x \leq -\frac{1}{2} + 1/n$. Thus, the first two eigenfunctions of (5.6) are simply $\tilde{y}_n(x)$ and $\tilde{y}_{2n}(x)$ restricted to the interval $-\frac{1}{2} \leq x \leq -\frac{1}{2} + 1/n$. Consequently, $\tilde{\lambda}_{2n}/\tilde{\lambda}_n = \lambda_2/\lambda_1$. We now have $\mu_{2n,n}(\alpha) \leq \tilde{\lambda}_{2n}/\tilde{\lambda}_n = \lambda_2/\lambda_1$ and $\nu_{2n,n}(\alpha) \geq \tilde{\lambda}_{2n}/\tilde{\lambda}_n = \lambda_2/\lambda_1$. Hence, $\mu_{2n,n}(\alpha) \leq \mu(\alpha)$ and $\nu_{2n,n}(\alpha) \geq \nu(\alpha)$ follow by letting ϕ_0 minimize or maximize λ_2/λ_1 since $\lambda_2 = \lambda_2(\phi_0)$ and $\lambda_1 = \lambda_1(\phi_0)$ and ϕ_0 extremizes λ_2/λ_1 .

LEMMA 2. $\mu_{2n,n}(\alpha) \geq \mu(\alpha)$ and $\nu_{2n,n}(\alpha) \leq \nu(\alpha)$ for $0 < \alpha \leq 1$ and $n = 1, 2, \dots$.

Proof: The proof is by induction. We only consider maximizing λ_2/λ_1 , the case of minimization following similarly.

(i) Let $\tilde{\phi}(x)$ maximize λ_4/λ_2 and let z_0 be the interior zero of y_2 and z_1 the middle zero of y_4 (which has three interior zeroes). We can take $z_0 \leq z_1$, since reflecting $\tilde{\phi}$ about $x = 0$, which does not alter the eigenvalues, requires the eigenfunctions to be reflected about $x = 0$.

There exists a smallest $\tilde{\lambda}_4 \geq \lambda_4$ such that $\tilde{y}(x)$, the solution of the problem $y'' + \tilde{\lambda}_4 \tilde{\phi}(x)y = 0$, $\tilde{y}(-\frac{1}{2}) = 0$, $\tilde{y}'(-\frac{1}{2}) = y_4'(-\frac{1}{2})$ has the property that $\tilde{y}(z_0) = 0$. Consider the new problem $y'' + \lambda \tilde{\phi}(x)y = 0$, $y(-\frac{1}{2}) = y(z_0) = 0$. The first eigenvalue of the new problem is λ_2 with eigenfunction $y_2(x)$ (because y_2 has no interior zeroes for $-\frac{1}{2} < x < z_0$). The second eigenvalue of the new problem is $\tilde{\lambda}_4$ with eigenfunction $\tilde{y}(x)$ (because $\tilde{y}$ has one interior zero for $-\frac{1}{2} < x < z_0$).

The transformation $L(x) = (2/(1 + 2z_0))(x + \frac{1}{2}) - \frac{1}{2}$ maps the interval of the new problem from $(-\frac{1}{2}, z_0)$ to $(-\frac{1}{2}, \frac{1}{2})$ and does not alter the ratio of the eigenvalues. We now have the problem

$$(5.7) \quad y'' + \lambda \tilde{\phi}(L(x))y = 0, \quad y(-\frac{1}{2}) = y(\frac{1}{2}) = 0.$$

The ratio of the first two eigenvalues of (5.7) is $\tilde{\lambda}_4/\lambda_2$. By our construction, $\tilde{\lambda}_4 \geq \lambda_4$. We have $\tilde{\lambda}_4/\lambda_2 \leq \nu(\alpha)$ by the definition of $\nu(\alpha)$, since $\tilde{\phi}(L(x))$

belongs to P_α . Therefore, $\nu_{4,2}(\alpha) \leq \nu(\alpha)$ since λ_4/λ_2 was the maximum, and $\lambda_4/\lambda_2 \leq \tilde{\lambda}_4/\lambda_2$.

(ii) We now assume that $\nu_{2k,k}(\alpha) \leq \nu(\alpha)$ for some integer k . Let $\tilde{\phi}(x)$ maximize $\lambda_{2(k+1)}/\lambda_{k+1}$. Further, let z_0 be the first interior zero of y_{k+1} less than $\frac{1}{2}$, and z_1 the second interior zero of $y_{2(k+1)}$ less than $\frac{1}{2}$, where $y_{2(k+1)}$ and y_{k+1} are the eigenfunctions corresponding to the eigenvalues $\lambda_{2(k+1)}$ and λ_{k+1} , respectively.

If $z_0 \leq z_1$, there exists a smallest $\tilde{\lambda}_{2(k+1)} \geq \lambda_{2(k+1)}$ such that $\tilde{y}(x)$, the solution of $y'' + \tilde{\lambda}_{2(k+1)}\tilde{\phi}(x)y = 0$, $\tilde{y}(-\frac{1}{2}) = 0$, $\tilde{y}'(-\frac{1}{2}) = y'_{2(k+1)}(-\frac{1}{2})$, has the property that $\tilde{y}(z_0) = 0$. Consider the new problem $y'' + \lambda\phi y = 0$, $y(-\frac{1}{2}) = y(z_0) = 0$. The k -th eigenvalue of this problem is λ_k , with eigenfunction $y_{k+1}(x)$ (since y_{k+1} has $k-1$ interior zeroes for $-\frac{1}{2} < x < z_0$). The $2k$ -th eigenvalue is $\tilde{\lambda}_{2(k+1)}$, with eigenfunctions $\tilde{y}(x)$ (since $\tilde{y}$ has $2k-1$ interior zeroes for $-\frac{1}{2} < x < z_0$). Using the same $L(x)$ as in (i), with the new z_0 , we conclude that $\tilde{\lambda}_{2(k+1)}/\lambda_{k+1} \leq \nu_{2k,k}(\alpha)$. But, $\nu_{2(k+1),(k+1)}(\alpha) = \lambda_{2(k+1)}/\lambda_{k+1}$, so $\nu_{2(k+1),(k+1)}(\alpha) \leq \nu_{2k,k}(\alpha)$.

By the induction hypothesis, $\nu_{2k,k}(\alpha) \leq \nu(\alpha)$, and so $\nu_{2(k+1),(k+1)}(\alpha) \leq \nu(\alpha)$.

We consider now the case $z_1 < z_0$. There exists a smallest $\tilde{\lambda}_{2(k+1)} > \lambda_{2(k+1)}$ such that $\tilde{y}(x)$, the solution to $y'' + \tilde{\lambda}_{2(k+1)}\tilde{\phi}(x)y = 0$, $y(\frac{1}{2}) = 0$, $y'(\frac{1}{2}) = y'_{2(k+1)}(\frac{1}{2})$, has the property that $\tilde{y}(z_0) = 0$. Consider the new problem $y'' + \lambda\tilde{\phi}(x)y = 0$, $y(\frac{1}{2}) = y(z_0) = 0$. The first eigenvalue is λ_{k+1} , with eigenfunction $y_{k+1}(x)$, and the second eigenvalue is $\tilde{\lambda}_{2(k+1)}$, with eigenfunction $\tilde{y}(x)$. Constructing the appropriate $L(x)$, we conclude that $\tilde{\lambda}_{2(k+1)}/\lambda_{k+1} \leq \nu(\alpha)$ (without using the induction hypothesis). By construction, $\nu_{2(k+1),(k+1)}(\alpha) = \lambda_{2(k+1)}/\lambda_{k+1} \leq \tilde{\lambda}_{2(k+1)}/\lambda_{k+1}$, and so $\nu_{2(k+1),(k+1)}(\alpha) \leq \nu(\alpha)$.

We have thus proved the identities $\nu_{2n,n}(\alpha) = \nu(\alpha)$ and $\mu_{2n,n}(\alpha) = \mu(\alpha)$ for $0 < \alpha \leq 1$, $n = 1, 2, \dots$.

Before proceeding further, we must state some results from Keller's paper which are derived using a variational analysis. Let $\phi_0(x)$ be a function in P_α which extremizes λ_2/λ_1 , with $y_1(x)$ and $y_2(x)$ the corresponding eigenfunctions which satisfy the normalization condition

$$\int_{-1/2}^{1/2} \phi_0(x) y_k^2(x) dx = 1, \quad k = 1, 2.$$

It is shown that

$$(5.8) \quad \phi_0(x) = \alpha \quad \text{or} \quad \phi_0(x) = 1 \quad \text{if} \quad \phi_0 \quad \text{is continuous at} \quad x.$$

Further, if ϕ_0 maximizes λ_2/λ_1 , we have

$$(5.9) \quad |y_1(x)| \leq |y_2(x)| \quad \text{if} \quad \phi_0(x) = \alpha,$$

$$(5.10) \quad |y_1(x)| = |y_2(x)| \quad \text{if} \quad \phi_0 \quad \text{has a jump at} \quad x,$$

$$(5.11) \quad |y_1(x)| \geq |y_2(x)| \quad \text{if} \quad \phi_0(x) = 1.$$

The inequalities in (5.9) and (5.11) are reversed if ϕ_0 minimizes λ_2/λ_1 , while (5.10) is unchanged. The same results hold for $y_j(x)$ and $y_k(x)$ if λ_j/λ_k is extremized, provided $k > j$, k replaces 2, and j replaces 1.

LEMMA 3. *Any function $\phi_0(x)$ which extremizes λ_2/λ_1 cannot have precisely one jump.*

Proof: Assume $\phi_0(x)$ extremizes λ_2/λ_1 and has just one jump. Then, reflecting about $x = 0$ if necessary, we have

$$(5.12) \quad \begin{aligned} \phi_0(x) &= \alpha, & -\frac{1}{2} \leq x < x_0, \\ \phi_0(x) &= 1, & x_0 < x \leq \frac{1}{2}. \end{aligned}$$

If ϕ_0 maximizes λ_2/λ_1 , (5.9) and (5.11) imply that $|y'_2(-\frac{1}{2})| \geq |y'_1(-\frac{1}{2})|$ and $|y'_2(\frac{1}{2})| \leq |y'_1(\frac{1}{2})|$. We maximize λ_4/λ_2 by constructing $\phi_2(x)$ from $\phi_0(x)$ as in (3.1). That is,

$$(5.13) \quad \begin{aligned} \phi_2(x) &= \alpha, & -\frac{1}{2} \leq x < \frac{1}{2}x_0 - \frac{1}{4}, \\ \phi_2(x) &= 1, & \frac{1}{2}x_0 - \frac{1}{4} \leq x < 0, \end{aligned}$$

and continue $\phi_2(x)$ as a periodic function, with period $\frac{1}{2}$, to $-\frac{1}{2} \leq x \leq \frac{1}{2}$. We know that ϕ_2 maximizes λ_4/λ_2 because $\nu_{4,2}(\alpha) = \nu(\alpha) = \lambda_2(\phi_0)/\lambda_1(\phi_0) = \lambda_4(\phi_2)/\lambda_2(\phi_2)$ by Lemmas 1 and 2 and the definitions of ϕ_0 and ϕ_2 .

From our construction of $\tilde{y}_2$ and $\tilde{y}_4$ in Lemma 1, we have

$$(5.14) \quad \tilde{y}_2(0) = \tilde{y}_4(0) = 0.$$

Without loss of generality, we can assume that $\tilde{y}'_2(0) > 0$ and $\tilde{y}'_4(0) > 0$ (multiplication by -1 will achieve this). Using (5.9), (5.11) and (5.13) we conclude that $\tilde{y}'_4(0^-) \leq \tilde{y}'_2(0^-)$ and $\tilde{y}'_4(0^+) \geq \tilde{y}'_2(0^+)$. Since $\tilde{y}_2$ and $\tilde{y}_4$ are C' functions, we see that

$$(5.15) \quad \tilde{y}'_4(0) = \tilde{y}'_2(0).$$

Further, from (5.9), there exists $\varepsilon > 0$ such that

$$(5.16) \quad \tilde{y}_4(x) \geq \tilde{y}_2(x) > 0, \quad 0 < x < \varepsilon.$$

By (1.1),

$$(5.17) \quad \tilde{y}_2'' - \tilde{y}_4'' = -\phi_2[\lambda_2(\phi)\tilde{y}_2 - \lambda_4(\phi_2)\tilde{y}_4].$$

Integrating (5.17) between 0 and x , with $x < \varepsilon$, using (5.15), (5.16), and the fact that $\lambda_4(\phi_2) > \lambda_2(\phi_2)$ (since the eigenvalues are simple), we have

$$(5.18) \quad \begin{aligned} \tilde{y}'_2(x) - \tilde{y}'_4(x) &= - \int_0^x \phi_2(s) [\tilde{\lambda}_2 \tilde{y}_2(s) - \tilde{\lambda}_4 \tilde{y}_4(s)] ds \\ &\equiv H(x) > 0, \end{aligned} \quad 0 < x < \varepsilon,$$

where $\tilde{\lambda}_2 \equiv \lambda_2(\phi_2)$ and $\tilde{\lambda}_4 \equiv \lambda_4(\phi_2)$. Integrating (5.18) and using (5.14), we have

$$(5.19) \quad \tilde{y}_2(x) - \tilde{y}_4(x) = \int_0^x H(s) ds > 0, \quad 0 < x < \varepsilon.$$

Thus, $\tilde{y}_2(x) > \tilde{y}_4(x)$ for $0 < x < \varepsilon$, contradicting (5.16). Therefore, the assumption that the maximizing function for λ_2/λ_1 has just one jump leads to a contradiction. A similar argument works for the minimizing function, and so we see that the extremizing function for λ_2/λ_1 cannot have just one jump.

The proof of Theorem 1 will be complete if we show that the extremizing function cannot have more than two jumps. This result was also necessary to show that $\mu(\alpha)$ and $\nu(\alpha)$ actually obtain their extreme values in P_α .

LEMMA 4. *If ϕ_0 extremizes λ_2/λ_4 , then $|y_1(x)|$ and $|y_2(x)|$ have at most two intersections in $(-\frac{1}{2}, \frac{1}{2})$, and ϕ_0 has at most two jumps there.*

Proof: By (5.10), it suffices to prove that $|y_1|$ and $|y_2|$ have at most two intersections on $(-\frac{1}{2}, \frac{1}{2})$. Let $\bar{x}$ be the interior zero of $y_2(x)$. We show that $|y_1|$ and $|y_2|$ have at most one intersection in $(-\frac{1}{2}, \bar{x})$ and $(\bar{x}, \frac{1}{2})$. Since $y_2(\bar{x}) = 0$ and $y_1(\bar{x}) \neq 0$ (y_1 has no interior zeroes), $|y_1(\bar{x})| \neq |y_2(\bar{x})|$. We take $y_1(x) > 0$ on $(-\frac{1}{2}, \frac{1}{2})$ and $y_2(x) > 0$ on $(-\frac{1}{2}, \bar{x})$, with $y_2(x) < 0$ on $(\bar{x}, \frac{1}{2})$.

Assume $|y_1|$ and $|y_2|$ have at least two intersections in $(-\frac{1}{2}, \bar{x})$, and let x_0 and x_1 be the first two such points greater than $-\frac{1}{2}$, so $-\frac{1}{2} < x_0 < x_1$ and any other intersections lie in $(x_1, \bar{x})$. Thus, we have

$$(5.20) \quad y_1(x_0) = y_2(x_0), \quad y_1(x_1) = y_2(x_1).$$

We define $v(x) = y_2(x)/y_1(x)$ and note that $v(x_0) = v(x_1) = 1$. By the definition of $v(x)$, we have

$$(5.21) \quad v'(x) = \frac{y'_2 y_1 - y_2 y'_1}{y_1^2} = \frac{w}{y_1^2},$$

$$(5.22) \quad w \equiv y'_2 y_1 - y_2 y'_1.$$

Note that $w(-\frac{1}{2}) = 0$. Using (1.1), it follows that

$$(5.23) \quad w'(x) = y_1(x)y_2(x)\phi_0(x)[\lambda_1 - \lambda_2].$$

Hence, $w'(x) < 0$ for $\frac{1}{2} < x < \bar{x}$, and so $w(x) < 0$ for $-\frac{1}{2} < x \leq \bar{x}$. Therefore, $v'(x) < 0$ for $-\frac{1}{2} < x < \bar{x}$. But we have above $v(x_0) = v(x_1) = 1$, contradicting $v'(x) < 0$ on $(-\frac{1}{2}, \bar{x})$. Thus, $|y_1|$ and $|y_2|$ have at most one intersection on $(-\frac{1}{2}, \bar{x})$. Similarly, $|y_1|$ and $|y_2|$ have at most one intersection on $(\bar{x}, \frac{1}{2})$.

This completes the proof of Theorem 1.

THEOREM 2. *Any extremizing function is symmetric about $x = 0$.*

Proof: In order to prove this theorem we need another lemma.

LEMMA 5. *The transmission coefficients s and t for y_1 and y_2 , defined in the proof of Lemma 1, are equal when ϕ_0 extremizes λ_2/λ_1 .*

Proof: If ϕ_0 has no jumps, then $\phi_0 \equiv \text{constant}$ and the transmission coefficients both equal unity. Now let ϕ_0 have a jump point at x_0 . Construct $\phi_2(x)$, $\tilde{y}_2(x)$ and $\tilde{y}_4(x)$ as in (3.1), (5.3) and (5.4). The function $\phi_2(x)$ has jump points at $-\frac{1}{2} + \frac{1}{2}(x_0 + \frac{1}{2})$ and $\frac{1}{2}(x_0 + \frac{1}{2})$. It follows from (5.10) that

$$(5.24) \quad |\tilde{y}_2(-\frac{1}{2} + \frac{1}{2}(x_0 + \frac{1}{2}))| = |\tilde{y}_4(-\frac{1}{2} + \frac{1}{2}(x_0 + \frac{1}{2}))|,$$

$$(5.25) \quad |\tilde{y}_2(\frac{1}{2}(x_0 + \frac{1}{2}))| = |\tilde{y}_4(\frac{1}{2}(x_0 + \frac{1}{2}))|.$$

We now use (5.3) and (5.4) in (5.24), and find

$$(5.26) \quad |M_1 y_1(x_0)| = |M_2 y_2(x_0)|.$$

We have $|y_1(x_0)| = |y_2(x_0)|$ by (5.10), so

$$(5.27) \quad |M_1| = |M_2|.$$

Thus, the normalization constants appearing in (5.3) and (5.4) are equal. Substituting (5.3), (5.4) and (5.27) into (5.25), we find

$$(5.28) \quad |-M_1 s y_1(x_0)| = |M_2 t y_2(x_0)|.$$

Since $|M_1| = |M_2|$, $|y_1(x_0)| = |y_2(x_0)|$ and s and t are positive, we have

$$(5.29) \quad s = t.$$

The proof of Lemma 5 is now complete.

We note that if ϕ_0 is a constant, it is symmetric about $x=0$. Thus, to prove Theorem 2 we need only consider the case where ϕ_0 is not a constant. By Theorem 1, ϕ_0 has two jumps, at x_0 and x_1 , with $-\frac{1}{2} < x_0 < x_1 < \frac{1}{2}$. Consider the problem

$$(5.30) \quad y'' + \lambda \phi_0(x)y = 0, \quad -\frac{1}{2} < x < \frac{1}{2},$$

where $\phi_0(x)$ is the same constant in $(-\frac{1}{2}, x_0)$ and $(x_1, \frac{1}{2})$. Let $w_1(x, \lambda)$ and $w_2(x, \lambda)$ be solutions to (5.30) such that

$$(5.31) \quad w_1(-\frac{1}{2}, \lambda) = 0, \quad w_1'(-\frac{1}{2}, \lambda) = A,$$

$$(5.32) \quad w_2(\frac{1}{2}, \lambda) = 0, \quad w_2'(\frac{1}{2}, \lambda) = A,$$

where A is an arbitrary real number. Since $\phi_0(x)$ equals the same constant for $-\frac{1}{2} \leq x < x_0$ and for $x_1 < x \leq \frac{1}{2}$, we must have $w_1(x, \lambda) = w_2(-x, \lambda)$ for $x \in [-\frac{1}{2}, x_0]$ and $-x \in [x_1, \frac{1}{2}]$ (the intervals are closed at the x_0 and x_1 end points by continuity). Thus, $w_1(x_0, \lambda) = w_2(-x_0, \lambda)$ if $|x_0| \geq |x_1|$, and $w_1(-x_1, \lambda) = w_2(x_1, \lambda)$ if $|x_1| \geq |x_0|$.

At the jumps of ϕ_0 , (5.10) requires $|y_1(x_0)| = |y_2(x_0)|$ and $|y_1(x_1)| = |y_2(x_1)|$. Let y_1 and y_2 have common transmission coefficient β (they are equal by Lemma 5). By definition of the transmission coefficients, $-\beta y_1'(\frac{1}{2}) = y_1'(-\frac{1}{2})$ and $\beta y_2'(\frac{1}{2}) = y_2'(-\frac{1}{2})$. Using the above remarks, first with $\lambda = \lambda_1$ and then with $\lambda = \lambda_2$, and taking $|x_0| > |x_1|$, we find

$$(5.33) \quad |y_1(x_0)| = \beta |y_1(-x_0)|, \quad |y_2(x_0)| = \beta |y_2(-x_0)|.$$

But, $|y_1(x_0)| = |y_2(x_0)|$, so that $\beta |y_1(-x_0)| = \beta |y_2(-x_0)|$, or $|y_1(-x_0)| = |y_2(-x_0)|$. Since $|x_0| > |x_1|$, $x_0 \neq 0$ and thus $-x_0 \neq x_0$. We now have three points of intersection of $|y_1|$ and $|y_2|$ in $(-\frac{1}{2}, \frac{1}{2})$, contrary to Lemma 4. The assumption $|x_1| > |x_0|$ leads to a similar contradiction, and hence we must have $|x_0| = |x_1|$. We know that $x_0 \neq x_1$, so we see that

$$(5.34) \quad x_0 = -x_1.$$

Thus, $\phi_0(x)$ is symmetric about $x=0$ since it can only take two different values, and its jumps are symmetrically located about $x=0$.

This completes the proof of Theorem 2.

Theorems 1 and 2 show that the extremizing function for λ_2/λ_1 is either a constant or has two jumps and is symmetric about $x=0$. If ϕ_0 is a constant, then by (5.9) and (5.11), we have $\phi_0 \equiv 1$ or $\phi_0 \equiv \alpha$. In either case, $\lambda_2/\lambda_1 = 4$. We also know that $\mu(\alpha) \leq 4 \leq \nu(\alpha)$ for $0 < \alpha \leq 1$. Thus, in order to determine $\mu(\alpha)$ or $\nu(\alpha)$, we can assume that ϕ_0 has two jumps.

THEOREM 3. *If ϕ_0 extremizes λ_2/λ_1 , then ϕ_n extremizes λ_{2n}/λ_n in the same sense.*

Proof: By Lemma 1, $\lambda_{2n}(\phi_n)/\lambda_n(\phi_n) = \lambda_2(\phi_0)/\lambda_1(\phi_0)$. But $\nu_{2n,n}(\alpha) = \nu(\alpha)$ and $\mu_{2n,n}(\alpha) = \mu(\alpha)$. Choosing ϕ_0 first to maximize, and then to minimize, λ_2/λ_1 , we find

$$\frac{\lambda_{2n}(\phi_n)}{\lambda_n(\phi_n)} = \frac{\lambda_2(\phi_0)}{\lambda_1(\phi_0)} = \nu(\alpha) = \nu_{2n,n}(\alpha)$$

and

$$\frac{\lambda_{2n}(\phi_n)}{\lambda_n(\phi_n)} = \frac{\lambda_2(\phi_0)}{\lambda_1(\phi_0)} = \mu(\alpha) = \mu_{2n,n}(\alpha).$$

Thus,

$$\nu_{2n,n}(\alpha) = \frac{\lambda_{2n}(\phi_n)}{\lambda_n(\phi_n)} \quad \text{if } \phi_0 \text{ maximizes } \frac{\lambda_2}{\lambda_1},$$

and

$$\mu_{2n,n}(\alpha) = \frac{\lambda_{2n}(\phi_n)}{\lambda_n(\phi_n)} \quad \text{if } \phi_0 \text{ minimizes } \frac{\lambda_2}{\lambda_1}.$$

Bibliography

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